CSC542 Project First Submission: AI Hype Cycle Detector

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Introduction

Project Description

This project investigates the "AI Hype Cycle" by analyzing YouTube discourse patterns surrounding five major generative AI technologies: ChatGPT, GPT-4, Gemini, GitHub Copilot, and Sora. Gartner's Hype Cycle framework proposes that new technologies follow a predictable five-phase trajectory: Innovation Trigger, Peak of Inflated Expectations, Trough of Disillusionment, Slope of Enlightenment, and Plateau of Productivity. While widely cited in technology forecasting, empirical validation of this framework using real-world social media data remains limited. This project uses YouTube engagement metrics and video transcript sentiment as proxies for public hype intensity, testing whether this cyclical pattern manifests consistently across multiple AI technologies launched between 2021 and 2024.

Research Questions

This project addresses three primary questions:

1. Can YouTube discourse patterns be used to empirically identify the five natural phrases of the Gartner Hype Cycle?
2. To what extent can a technology's "Hype Phase" be accurately predicted using supervised classification models?
3. Do different AI technologies exhibit unique "discourse signatures," or do they follow universal trajectory through the Hype Cycle?

Justification

Validating the hype cycle empirically helps product teams (optimizing launch timing and communication strategies), investors (informing entry/exit decisions), and policymakers (improving risk communication and timing educational interventions) understand the predictable trajectory of AI adoption and public disillusionment.

State-of-the-Art

Sentiment Analysis

Current research in technology sentiment analysis predominantly relies on binary classification and is heavily built upon pre-2022 Twitter data. These models often fail to capture the nuanced technical discourse of the 'Generative AI era' and treat sentiment as a static snapshot rather than a shifting temporal state. However, recent studies, such as the analysis of public reactions to Sora, demonstrate the move toward fine-grained emotion detection and topic modeling to understand public perception during initial release windows (Yuan et al., 2024).. While that study explores the what of public emotion, it does not utilize Large Language Models (LLMs) for automated sentiment extraction. This project bridges that gap by using Gemini 2.5 Flash Lite to model sentiment shifts as sequential, predictable transitions.

Automating Technology Assessment

Conventional methods for assessing technology maturity, such as the S-curve, Technology Readiness Levels (TRL), and Gartner's Hype Cycle, have historically depended on subjective expert opinions. Recent scholarship highlights the limitations of these manual approaches and advocates for a transition to automated technology maturity assessment using big text data and machine learning (Kuznetsov et al., 2017). Our project directly addresses this by using Hierarchical Clustering and Random Forest models to validate that Gartner's theoretical phases can be empirically identified through Youtube discourse.

Empirical Validation of the Hype Cycle

While longitudinal studies have been conducted on older technologies to compare Gartner's reported progression against real-world data (Dedehayir & Steinert, 2016), there is a lack of similar empirical scrutiny for modern Generative AI. Our work expands on existing methods by normalizing engagement data against channel-level metadata to track the life cycle of AI technologies across different launch years (2021-2024).

Novelty

This project's novelty lies in four key areas that advance the current state-of-the-art:

1. **Multi-Technology Comparative Framework:** We conduct a cross-sectional comparison of ChatGPT, GPT-4, Gemini Copilot, and Sora. This allows us to test pattern consistency across the AI landscape, proving that the Hype Cycle is a universal framework, technologies like Sora (Peak) and ChatGPT (Plateau) move through it at distinct paces.
2. **LLM-Driven Discourse Feature Extraction:** Unlike traditional lexicon-based approaches, we utilize Gemini 2.5 Flash Lite to extract complex features: Sentiment, Hyperbole, and Technical Complexity. This enables a more nuanced characterization of the "vibe" of technical discourse.
3. **Platform-Native Proxy Metrics:** We utilize YouTube engagement metrics (view-to-like ratios and comment density) as proxies for hype intensity, filling a significant gap in how video dynamics are used to map product maturity.
4. **High-Accuracy Predictive Modeling:** Beyond empirical observation, we demonstrate that these theoretical Hype phases are statistically distinct ($p<0.001$) and highly predictable through machine learning. Our Linear Support Vector Machine (SVM) model achieved a 92.3% test accuracy, significantly outperforming baseline models. This proves that the "discourse signatures" of hype are linearly separable and distinct enough to allow for near-perfect automated classification of technology maturity.

Materials

Data Collection

Data were collected programmatically via the YouTube Data API v3 and the youtube-transcript-api Python library across five generative AI technologies: ChatGPT, GPT-4, Gemini, GitHub Copilot, and Sora. No pre-existing dataset was downloaded; all records were generated through targeted API queries. Data collection was distributed across three team members for API efficiency, each responsible for a subset of technologies, with outputs subsequently merged into a single dataset.

Collection Design

For each technology, a 78-week window was constructed (26 pre-launch, 52 post-launch), divided into 39 bi-weekly periods. Within each window, up to 80 candidate videos were retrieved by querying the API with technology-specific search terms ordered by view count. After applying inclusion criteria, the 15 highest-viewed survivors per window were retained along with their metadata, channel statistics, and transcripts. We acknowledge that view-ranked sampling may underrepresent highly technical content, which tends to attract smaller audiences despite substantive impact; to partially address this, technical complexity was explicitly modeled as a separate feature.

Inclusion Criteria

Videos were retained only if they met all three conditions: a minimum of 1,000 views, a minimum duration of 120 seconds, and a primary or default audio language of English. Note that full transcripts were retrieved where available; for videos lacking manual captions, auto-generated English transcripts were used as a fallback.

Technology Assignment

To ensure each video was correctly assigned to its target technology, a weighted keyword-matching heuristic was applied across title (3×), description (2×), and transcript (1×). Each technology was defined by a specific set of keywords and synonyms to capture varied naming conventions; for example, GPT-4 included terms like "gpt-4o," "gpt-4 turbo," and "gpt4v," while Gemini included "google bard" and "gemini pro." Each video was assigned strictly to the technology with the *highest* cumulative weighted score across these fields. Videos were discarded if this highest-scoring assignment differed from the initial search query; this was essential for isolating specific model versions, such as when a search for GPT-4 returned results where GPT-3.5 was actually the dominant topic of the content. Additionally, videos were discarded where all keyword scores across technologies were zero. This typically occurred when YouTube's search returned videos based on irrelevant tags from the creator, like "ChatGPT", even though the technology was never actually mentioned in the title, description, or transcript.

LLM Feature Extraction

As illustrated in the data collection framework (Figure 1), each remaining video's title, description, and transcript were submitted to Gemini 2.5 Flash Lite (temperature = 0) with a structured prompt including scoring rubrics and labeled boundary examples. Six features were extracted per video:

- **Sentiment (−1 to 1):** overall tone toward the technology, from catastrophic/alarmist to euphoric. Technical vocabulary was explicitly defined as neutral in the prompt to avoid false negatives.
- **Hyperbole (0–1):** degree of language exaggeration independent of sentiment, from measured and academic to maximally exaggerated.

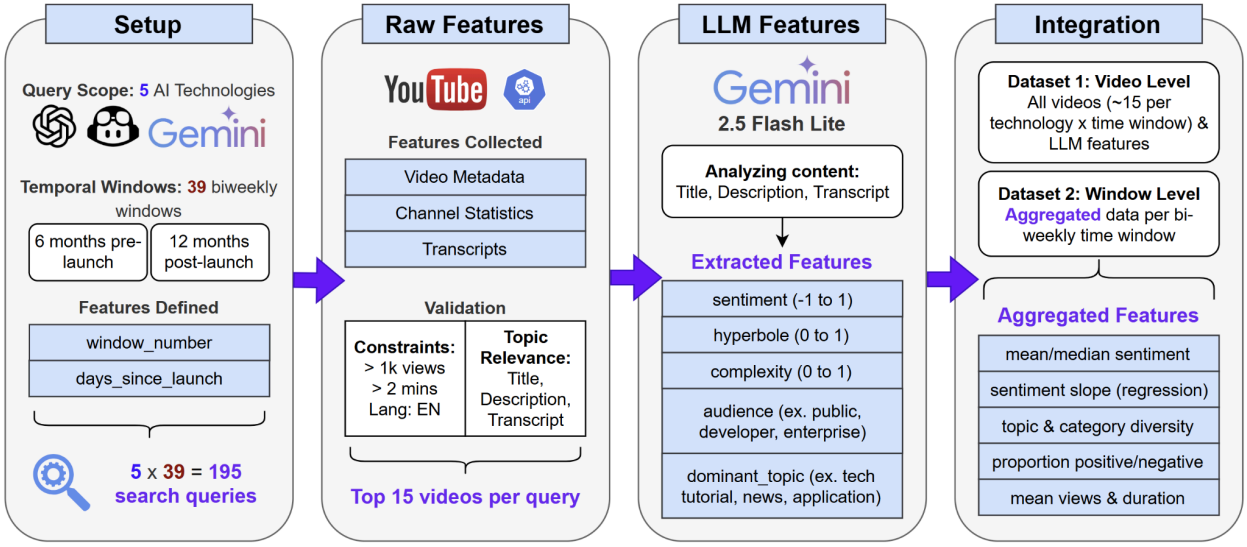
- **Complexity (0–1):** technical depth of the content, from general-public analogies to expert-level discussion of model internals, fine-tuning, and benchmarks.
- **Audience:** exactly one of General Public, Developers/Technical, or Business/Enterprise.
- **Dominant topic:** exactly one of Enthusiast/Reaction, Technical Tutorial, Ethics/Safety, News/Analysis, Critical/Evaluative, or Practical Application.
- **Confidence (0–1):** reliability of the above scores given transcript quality and topical relevance.

Videos with a confidence score below 0.3 were excluded as off-topic or transcript-deficient. This primarily filtered out ambiguous search results; for example, SoraAI queries returned content generated with Sora, such as music, that carried no transcript signal for the target technology.

Aggregation

Video-level data were aggregated to 158 technology × bi-weekly window observations. Window-level features included mean/median sentiment, sentiment variability, proportion positive/negative sentiment, mean hyperbole, mean complexity, like and comment ratios (per view), topic and category diversity, proportion of Science & Technology videos, mean duration, mean subscriber count, and a lagged sentiment slope computed via linear regression over the preceding five windows.

Figure 1. Schematic of the data collection framework



Methods

The analytical pipeline proceeded in two distinct phases: the unsupervised discovery of discourse-based "hype phases" followed by the supervised classification of those phases to evaluate their predictability.

Unsupervised Clustering

Feature Selection and Bias Mitigation
Building on the extraction process in Figure 1, the analytical dataset comprised 24 window-level variables. These included the six LLM-extracted semantic features (Sentiment, Hyperbole, Complexity, Audience Type Category, Dominant Topic Category, and Extraction Confidence) alongside YouTube API-derived metadata such as duration, YouTube-identified video category, and channel/engagement statistics.

For the unsupervised discovery phase, we specifically excluded categorical variables and raw engagement totals. While one-hot encoded for downstream classification, categorical features were withheld from clustering to maintain the mathematical integrity of the distance metrics, which are optimized for continuous numerical data. Similarly, raw view and subscriber counts were omitted to mitigate popularity bias; since established tools like ChatGPT naturally command higher reach, their inclusion would have likely clustered observations by "fame" rather than discourse maturity. To identify these "hype signatures," hierarchical clustering was applied to eight scaled numerical features across four functional domains:

- **Linguistic (LLM-generated):** Mean Sentiment and Mean Hyperbole.
- **Content (LLM/Metadata):** Mean Complexity and the proportion of videos categorized by YouTube as "Science & Technology".
- **Diversity (Metadata):** Topic Diversity (normalized unique topics) and Category Diversity.
- **Engagement (Metadata):** Mean Like-to-View and Comment-to-View ratios.

Methodological Selection and Temporal Validation
To identify the most appropriate clustering architecture, we evaluated several configurations, including Euclidean-based k-means and Ward's linkage. These methods were assessed based on their ability to produce temporally distinct groupings without using days since launch as a clustering feature. This approach allowed us to test our hypothesis that specific discourse signatures form groupings that align with distinct points in a technology's maturity, a quality formally evaluated in the statistical validation section below. Ultimately, correlation-based distance with average linkage was selected because it prioritizes the relative pattern of discourse features rather than absolute magnitude, effectively isolating the "shape" of the signatures over time regardless of a technology's total view volume.

Statistical Validation of Hype Phases
The resulting five clusters (k=5) were reordered by their mean days since launch to represent a chronological progression from Phase 1 through Phase 5. This validation strategy relied on intentionally withholding of days since launch variable from the clustering algorithm to keep the model "blind" to the technology's age, allowing us to utilize temporal data as an independent ground truth to confirm whether these hype signatures naturally cluster in a way that is distinct temporally.

To verify these alignments, a One-Way ANOVA was utilized as the global test of significance. This was the appropriate choice for the data as it allowed for the simultaneous evaluation of means across all five

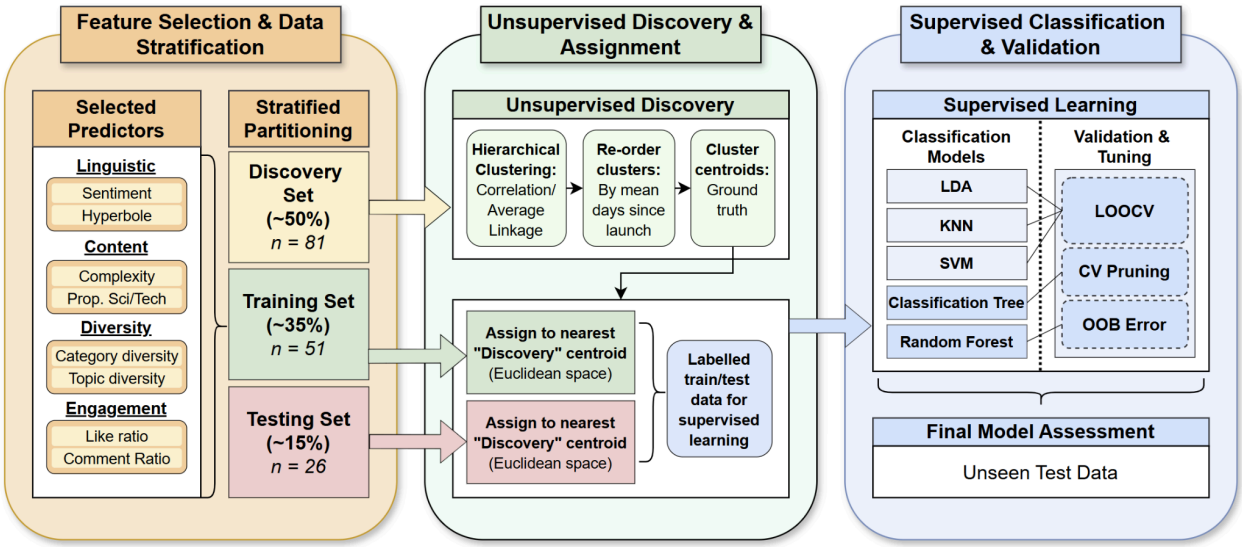
clusters to determine if "Hype Phase" (the categorical cluster assignment) was a statistically significant predictor of temporal maturity. This was further refined by a Tukey Honest Significance Difference (HSD) post-hoc test to pinpoint exactly where significant temporal shifts occurred between specific clusters. Tukey HSD was selected to adjust for the family-wise error rate, ensuring that any identified distinctions between phases were statistically robust and not the result of multiple-comparison chance. This blind validation procedure was designed to determine if the identified linguistic and engagement signatures serve as legitimate, time-dependent proxies for the theoretical phases of the Gartner Hype Cycle.

Supervised Classification

To determine whether the discovered Hype Cycle phases were representative of stable discourse signatures, we transitioned to a supervised learning framework. This stage was designed to assess if the patterns observed during the unsupervised discovery were generalizable and carried sufficient predictive signal to classify new observations. By attempting to predict cluster assignments from a separate set of features, we evaluated the degree to which the identified hype phases represent distinct, recognizable states in the public and technical consciousness rather than artifacts of the discovery partition.

As illustrated in the schematic of the supervised prediction pipeline (Figure 2), the analysis proceeded through three integrated stages: data stratification, unsupervised assignment, and supervised classification. The first stage utilized a stratified partitioning strategy to divide the data into three distinct sets to maintain strict experimental integrity. To ensure a completely unbiased final evaluation, a held-out Testing Set (15%, n=26) was first removed. The remaining 85% of the data was then divided using a 60/40 split, resulting in a Discovery Set (51%, n=81) used for cluster generation and a Training Set (34%, n=51) used for model development. This multi-stage partitioning ensured that the classification models were trained and tested on data entirely independent of the discovery process.

Figure 2. Schematic of the supervised prediction pipeline



In the second stage, Unsupervised Discovery and Assignment, the hierarchical clustering methodology - utilizing the same correlation-based distance and average linkage as the primary analysis - was re-performed exclusively on the Discovery Set to establish "ground truth" centroids. Observations in the Training and Testing sets were then assigned labels by mapping them to their nearest discovery centroid in Euclidean space. While correlation was utilized initially to define the qualitative shape and pattern of the hype signatures, the transition to Euclidean mapping was required to measure the geometric proximity

of new observations to the fixed coordinates of the established centroids in the standardized feature space. This step transformed the unsupervised clusters into a labeled dataset, enabling us to test if these signatures remained spatially consistent across novel observations.

The third stage, Supervised Classification and Validation, involved training and optimizing a suite of predictive models to evaluate the stability of the discovered signatures. During this phase, the predictor set was refined to maximize accuracy while ensuring model parsimony, a necessary constraint given the training sample size (n=51). We utilized the core linguistic and metadata features from the previous stages but replaced the aggregate topic diversity metric with two specific one-hot encoded dominant topics extracted during the LLM processing phase: “Practical Application” and “Critical/Evaluative.” These topics were selected after empirical performance testing demonstrated they served as the most effective semantic proxies for the theoretical Slope of Enlightenment and Trough of Disillusionment phases, respectively. To identify the optimal mathematical approach for the discourse space, we evaluated a diverse range of classifiers, including Linear Discriminant Analysis (LDA), K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and ensemble-based trees. Model parameters were optimized within the training partition using LOOCV for the parametric models and CV-based pruning or Out-of-Bag (OOB) error for the tree-based architectures. The final model assessment was conducted exclusively on the held-out testing set to determine the ultimate generalizability and predictive reliability of the discourse-based classification framework.

Results

Descriptive Analysis

The aggregated dataset consists of 158 time windows analyzed across five technologies and 24 variables. Missing values were restricted to sentiment_std (n=16) and sentiment_slope (n=15), as these metrics depend on previous time windows or multiple videos within a window to compute variance . Semantic analysis of the dominant topics revealed a heavy concentration in Practical Application (n=101) and Enthusiast/Reaction (n=40) content, accounting for the vast majority of discourse . In contrast, Critical/Evaluative (n=6), News/Analysis (n=7), and Technical Tutorial (n=4) windows were comparatively rare . Metadata categories were similarly dominated by Science & Technology (n=87) and Education (n=48), while remaining categories like Gaming and Entertainment appeared 10 or fewer times each .

Unsupervised Learning: Cluster Analysis

Hierarchical clustering using correlation-based distance and average linkage (k=5) was applied to the full 158-window dataset to identify natural groupings in public and technical discourse. The resulting clusters were reordered by their mean days since launch to represent a chronological progression from Phase 1 to Phase 5. Table 1 summarizes the key characteristics of these identified phases.

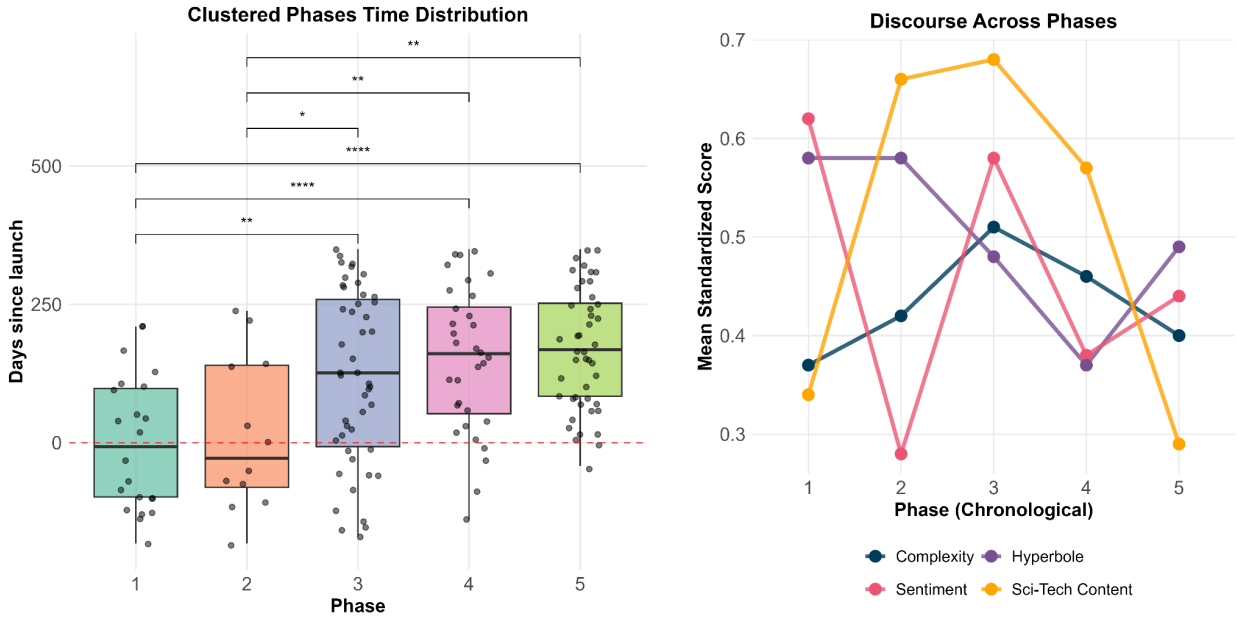
Table 1.

Cluster	n	Mean Days	Sentiment	Hyperbole	Complexity	Prop. Sci/Tech
1	22	0	0.62	0.58	0.37	0.34
2	12	14	0.28	0.58	0.42	0.66
3	47	118	0.58	0.48	0.51	0.68

4	32	149	0.38	0.37	0.46	0.57
5	45	166	0.44	0.49	0.40	0.29

As shown in Figure 3, these clusters align with the theoretical phases of the Gartner Hype Cycle. Cluster 1 (avg. 0 days) represents the Peak of Inflated Expectations, exhibiting the highest sentiment and hyperbole alongside the lowest technical complexity. Cluster 2 (avg. 14 days) aligns with the Trough of Disillusionment, where sentiment reaches its nadir but high science/tech proportions signal critical evaluation of technical realities. Clusters 3 and 4 (avg. 118–149 days) correspond to the Slope of Enlightenment, showing a recovery in sentiment and higher technical depth as early adopters accept the technology's capabilities . Finally, Cluster 5 (avg. 166 days) indicates entry into the Plateau of Productivity, with stable sentiment and hyperbole as the technology reaches a broader, mainstream audience.

Figure 3.

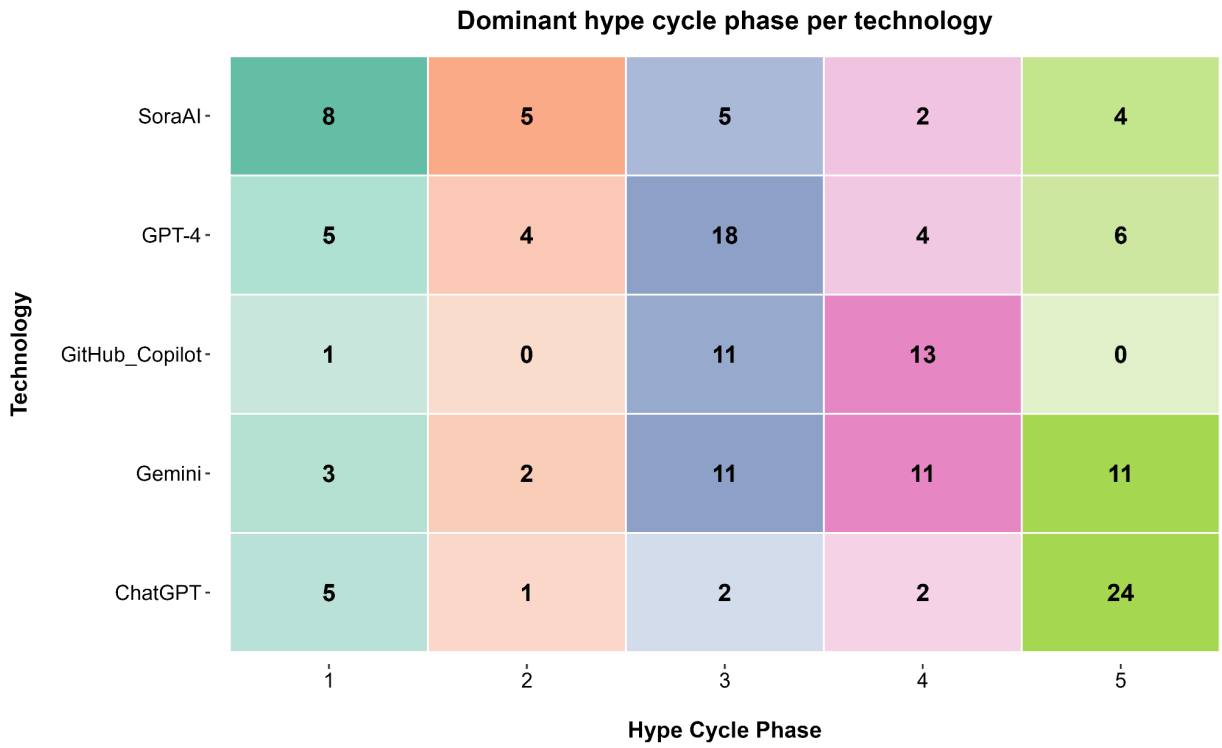


The statistical validity of these clusters was confirmed through a One-Way ANOVA, which demonstrated that "Hype Phase" is a significant predictor of days since launch ($F = 8.069$, $p = 6.26 \times 10^{-6}$). Because time was withheld from the clustering algorithm, this result provides independent validation that these discourse signatures are intrinsically tied to temporal maturity. Tukey HSD post-hoc tests further clarified the temporal separation between phases: Cluster 1 was significantly earlier than Clusters 3 ($p = 0.0068$), 4 ($p = 0.0007$), and 5 ($p < 0.0001$). Similarly, Cluster 2 was significantly distinct from the later stages in Cluster 4 ($p = 0.0259$) and Cluster 5 ($p = 0.0048$). The lack of significant differences between adjacent later clusters (e.g., 3 vs 4, $p = 0.844$) reflects the gradual nature of post-peak stabilization.

Per-technology phase distributions, shown in Figure 4, offer further interpretive grounding. Sora AI is concentrated in Cluster 1 (n=8 of 24 total windows), characterized by high sentiment and hyperbole with low complexity, which is consistent with a technology still in peak hype driven by viral visual demonstrations rather than practical use. GPT-4 predominantly falls in Cluster 3 (n=18 of 37 windows), suggesting a transition toward more measured technical evaluation. GitHub Copilot is heavily concentrated in Cluster 4 (n=13 of 25 windows, with zero windows in Cluster 5), which has low sentiment and the highest proportion of Science/Tech videos, indicating its developer audience has

surfaced implementation limitations characteristic of the Trough. Gemini is distributed broadly across the later phases of the cycle, with equal concentration in Clusters 3, 4, and 5 (n=11 each), reflecting sustained technical and practical scrutiny. ChatGPT overwhelmingly falls in Cluster 5 (n=24 of 34 windows), the furthest along the cycle of any technology, with discourse having shifted from speculation toward everyday utility.

Figure 4.



Supervised Learning: Phase Classification

To evaluate the generalizability of the identified discourse signatures, we transitioned to supervised learning by assigning ground-truth labels to the Training Set (n=51) and Testing Set (n=26) based on their proximity to the discovery centroids in Euclidean space. As shown in the performance summary below, we evaluated a diverse suite of classifiers using LOOCV, CV-pruning, and OOB error to optimize model parameters within the training partition.

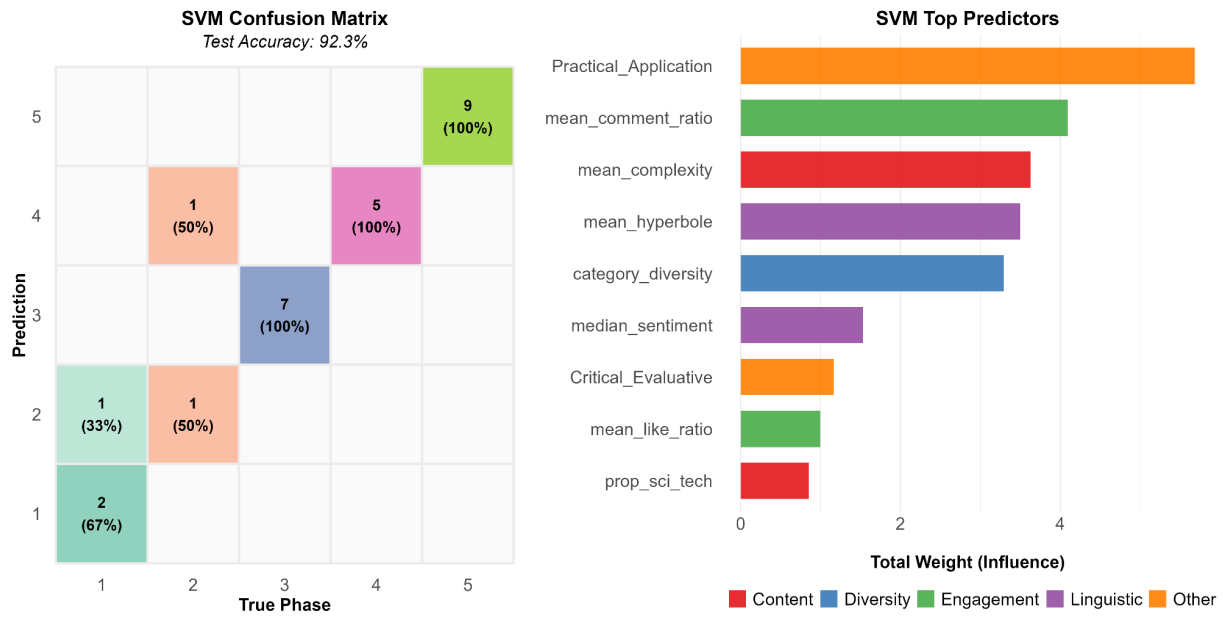
Table 2.

Model	Test Accuracy	CV Accuracy	CV Method
SVM (Linear Kernel)	**0.9231**	**0.8824**	**LOOCV (train)**
SVM (Radial Kernel)	0.8846	0.8627	LOOCV (train)
LDA	0.8462	0.8431	LOOCV (train)
KNN (K=5)	0.8077	0.7843	LOOCV (train)
Random Forest	0.7308	0.7451	OOB (train)

KNN (K=2, LOOCV-selected)	0.6923	0.7843	LOOCV (train)
KNN (K=1)	0.7308	0.6275	LOOCV (train)

The Linear SVM emerged as the superior architecture, achieving a 92.31% test accuracy. This high level of performance on the held-out testing set confirms that the identified "hype signatures" are not artifacts of the initial discovery partition but represent stable, recognizable patterns in public discourse. As illustrated in the SVM Confusion Matrix (Figure 6), the model successfully recovered the correct phase assignments for nearly all test observations, with minimal confusion occurring only between temporally adjacent clusters.

Figure 6.



Feature Importance (Linear SVM)

To understand the specific drivers of phase classification, we extracted and analyzed the weights from the top-performing Linear SVM model. As shown in the SVM Top Predictors analysis (Figure 6), the model's decisions are anchored by a combination of LLM-extracted semantic markers and API-derived engagement metrics.

The "Practical Application" topic emerged as the single most influential predictor by a significant margin, serving as the primary marker for the later, more mature phases of the cycle. This was followed by Engagement metrics, specifically the mean_comment_ratio, and Content features like mean_complexity. Linguistic markers, including mean_hyperbole and median_sentiment, also provided substantial predictive signal, particularly in identifying the high-affect "Peak" and "Trough" phases.

The high influence of the "Critical/Evaluative" topic further confirms the model's ability to distinguish the Trough of Disillusionment from other stages. Overall, these results indicate that the transition between hype phases is defined by a measurable shift from speculative, high-hyperbole news toward complex, practically-oriented content that invites deeper user engagement through comments.

Conclusions & Future Works

Conclusion

This study successfully demonstrated that the Gartner Hype Cycle is not merely a theoretical framework but an empirically detectable phenomenon within social discourse. By analyzing YouTube metadata and linguistic signatures through LLM-driven feature extraction, we achieved three primary objectives:

- **Empirical Validation:** Through empirical clustering and ANOVA testing ($P<0.001$), we proceed that linguistic patterns in AI-related discourse parallel the five stages of the Gartner Hype Cycle. These clusters represent statistically distinct phases in a technology's lifecycle, moving from viral, high-hyperbole triggers to stable, utility-focused discussions.
- **Reliable Predictive Modeling:** The transition from unsupervised discovery to supervised classification yielded exceptional results. Our Linear SVM model achieved a 92.3% test accuracy, proving that discourse features, specifically sentiment, hyperbole, and technical complexity, form a predictable and machine-learnable data signature for technology maturity.
- **Technological Mapping:** We successfully mapped current AI technologies onto the cycle. It was seen that while Sora remains concentrated at the "Peak of Inflated Expectations," ChatGPT has pioneered the transition into the "Plateau of Productivity," serving as a benchmark for AI maturity.

Future Works

The following directions could be used for extending this research:

1. **Objective Feature Selection:** The eight features used in clustering were selected based on theoretical reasoning, which introduces a degree of researcher bias. Future work should explore data-driven feature selection methods such as automated importance rankings or dimensionality reduction techniques.
2. **Validation Against Traditional Sentiment Methods:** This project relied on Gemini 2.5 Flash Lite to extract sentiment, hyperbole, and complexity scores. However, these LLM-derived features have not yet been cross-validated against established sentiment analysis tools such as VADER or LIWC. Comparing both approaches would help determine whether LLMs genuinely capture richer nuance in technical AI discourse, or whether traditional lexicon-based methods produce comparable results at lower computational cost.
3. **Evaluation of Alternative Technology Maturity Frameworks:** This study was designed specifically around Gartner's five-phase Hype Cycle. Future research could apply alternative frameworks, such as the S-curve diffusion model or Technology Readiness Levels (TRL), to the same dataset, allowing for a direct comparison of how well each theoretical model aligns with real-world discourse patterns observed on YouTube.
4. **Multi-Platform Expansion:** While YouTube provides a robust communal discourse, expanding the dataset to Reddit and X (formerly Twitter) would provide a richer, cross-community signal. Integrating these sources would help determine if "hype signatures" are universal or platform-dependent
5. **Advanced Modeling (PLS-DA):** To better handle the inherent correlation between features like sentiment and hyperbole, future iterations should utilize Partial Least Squares Discriminant Analysis (PLS-DA). Unlike traditional LDA, PLS-DA is specifically designed for high-dimensional, correlated feature spaces and could extract more discriminative latent components to further increase classification accuracy.

References

Dedehayir, O., & Steinert, M. (2016). The hype cycle model: A review and future directions. *Technological Forecasting and Social Change*.
Kuznetsov, E., et al. (2017). Prospects for Big Text Data Application in Technology Maturity Assessment. *IEEE Xplore / ResearchGate*.
Yuan, X., et al. (2024). Sora: A Two-Part Analysis of Initial Public Reaction to Text-to-Video AI. *MDPI Applied Sciences*.

Github Link

https://github.com/Korinne-Ciechon/CSC542-Final-Project-AI-Hype-Cycle